

SIMATS ENGINEERING

ASSESSMENT TOOL 1 (High)- Analytical Problem Solving

COURSE NAME: Data Structures

COURSE CODE:CSA03

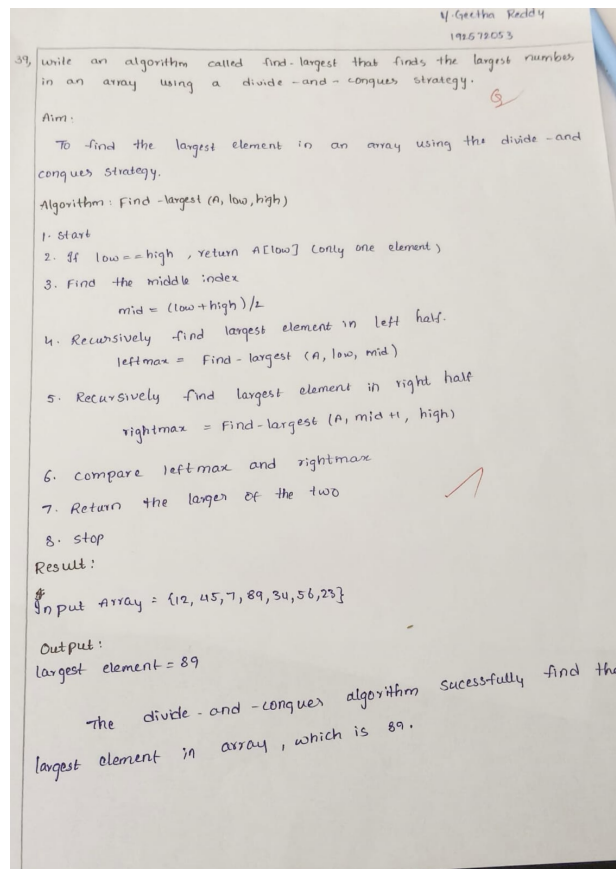
COURSE OUTCOME COVERED

C02 : Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. (BTL3, BTL4)

Assessment Tool Weightage: **30%**

QUESTIONS:

39. Write an algorithm called Find-Largest that finds the largest number in an array using a divide-and-conquer strategy.



40. Print all the prime integers present in the Single linked list having a list of 24->21->57->31->12->47->7->38->.

[illegible]

Code of Conduct:

I  (192572053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation